Agile Lean Six Sigma
Hacking
Lean Six Sigma for Results

Jay Arthur

March 25, 2021



Learning Objectives

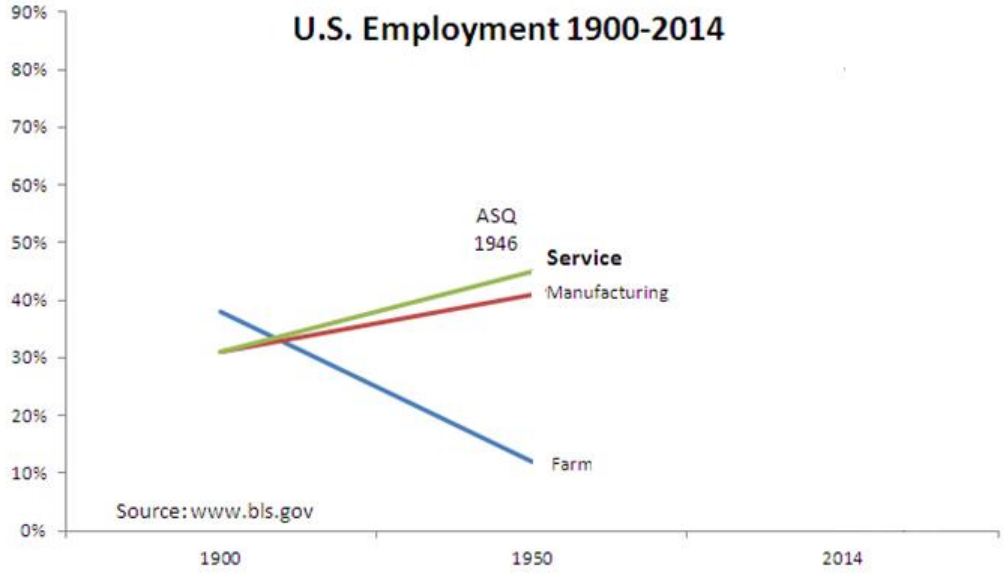
- Companies are stumbling onto this Agile approach to implementing Lean Six Sigma and succeeding where tradition implementations have failed to deliver results.
- Accelerate implementation of Lean Six Sigma and deliver results
- Identify best candidates for Green or Black Belt training
- Identify the seven key tools for Lean Six Sigma success
- Learn how to teach Lean Six Sigma in one day and generate results

Six Sigma is *too complex and time-consuming to fit into a regular workday. We need tools that don't require the entire organization to undergo weeks-long training programs.*

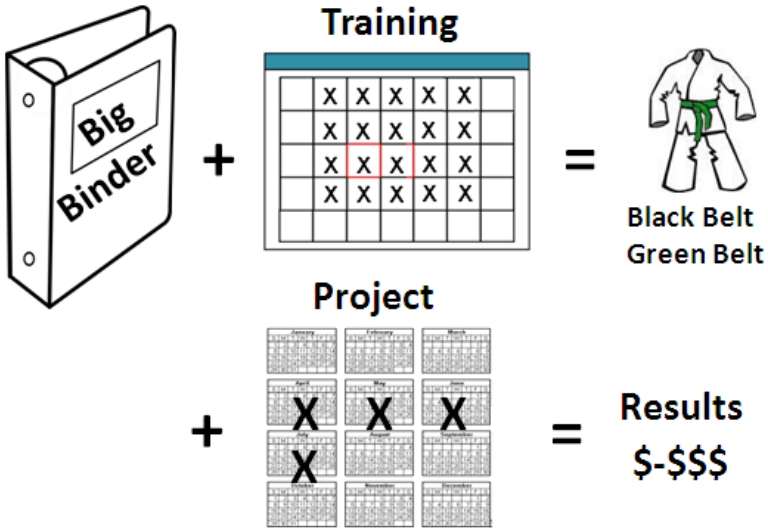
Thomas Wedell-Wedellsborg

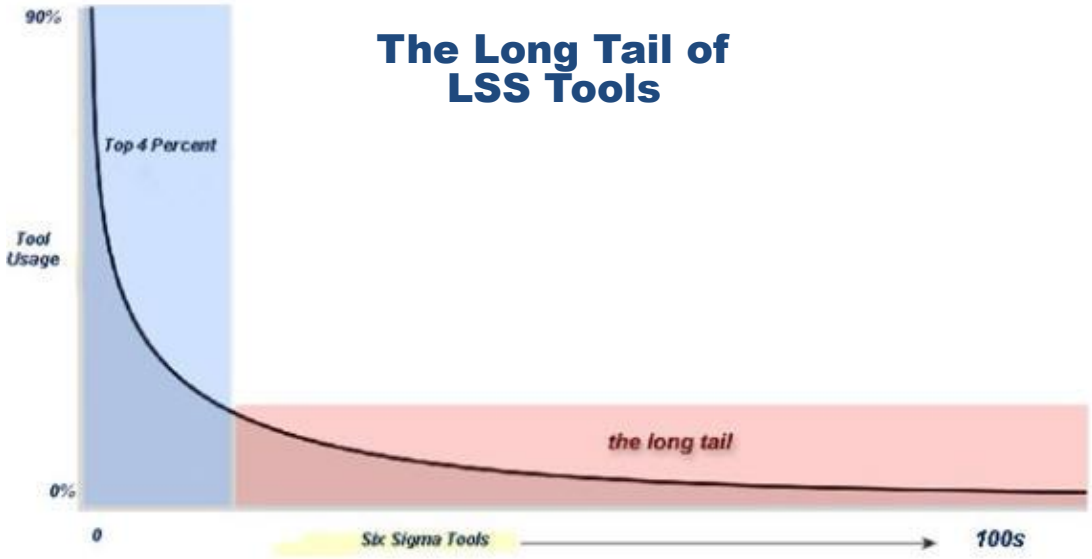
January-February 2017 HBR

20th Century Quality Improvement



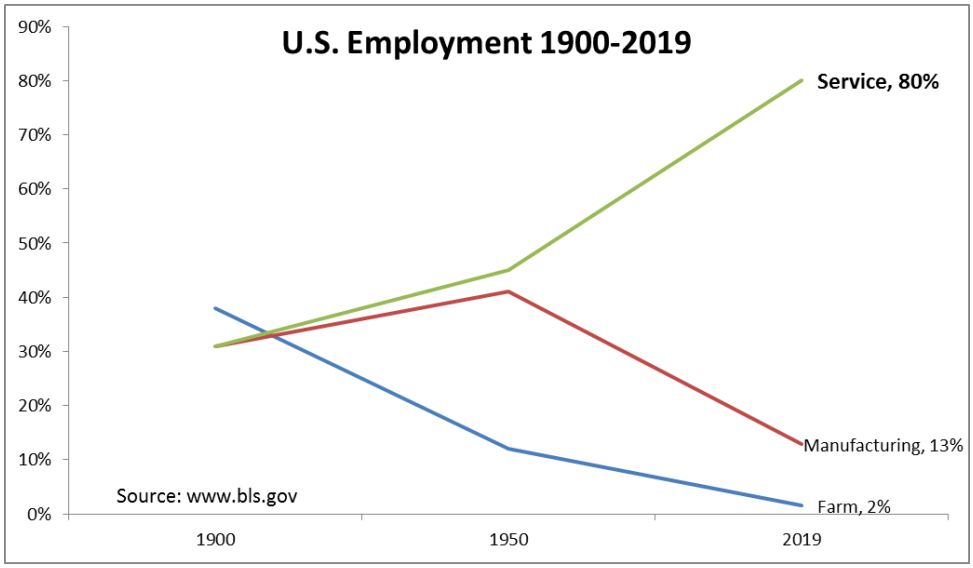
Traditional Lean Six Sigma





The Long Tail of LSS Tools

21th Century Quality Improvement



Problems	Methods	Tools
Complicated Problems	DMAIC, DOE	Complex
Uncomplicated Problems	Focus Improve Sustain Honor	Data + 7 Key Tools
Simple Problems	Trial & Error	None

Hacking Lean Six Sigma

The spirit of hacking can be adapted and applied to general business management, not just technical innovation.

Scott Brinker – Hacking Marketing

Agile Hacks Rediscovered









Hacking Lean Six Sigma for Results

Prework

Select teams that have a real problem to solve and data about the problem.

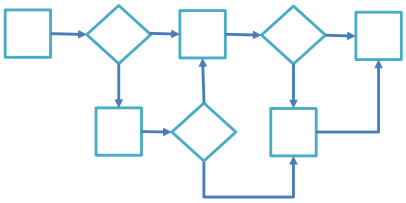
One-Day

Train teams in key improvement tools and apply to their data and problem.

Postwork

Implement countermeasures and measure results

Hacking Lean





Value Stream Mapping

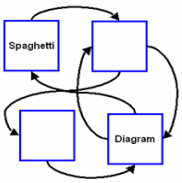


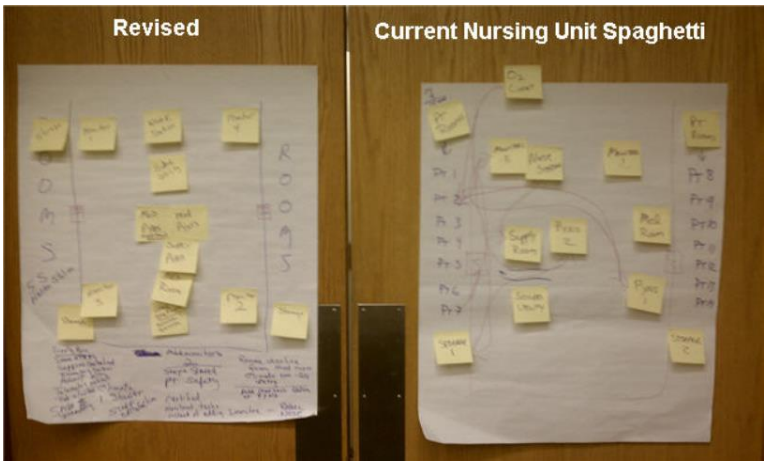




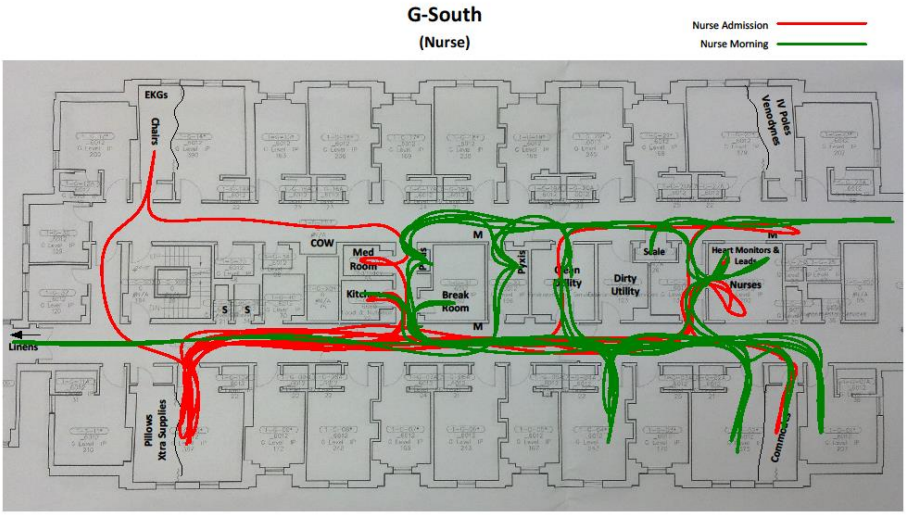


Spaghetti Diagramming

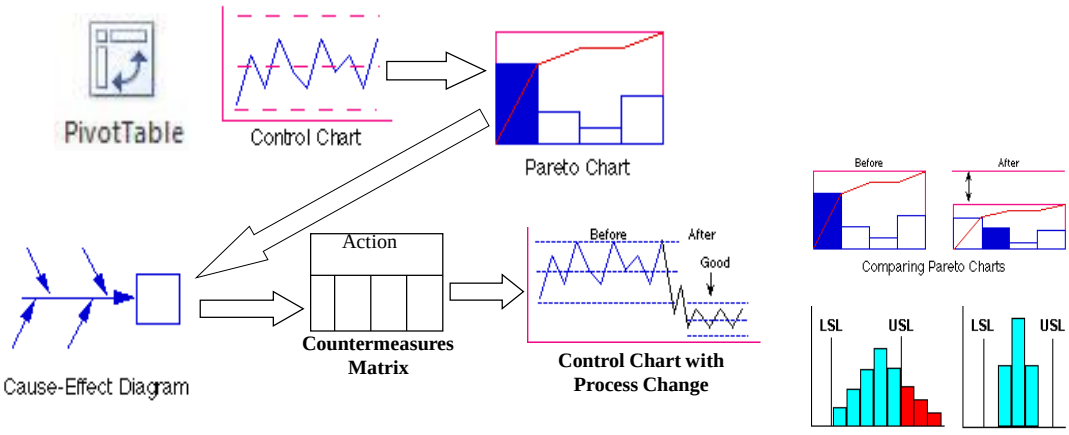




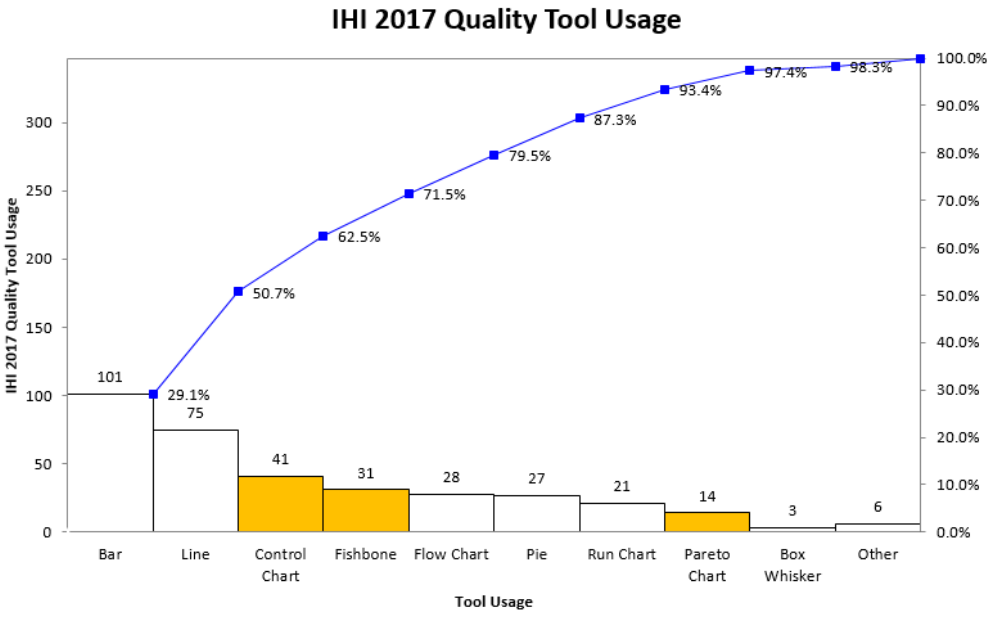
Nursing Unit Spaghetti Diagram

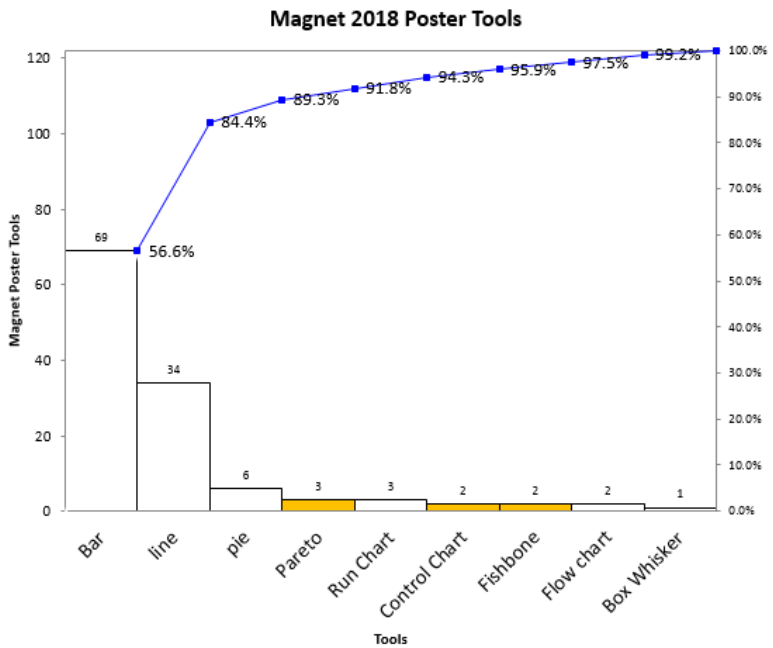


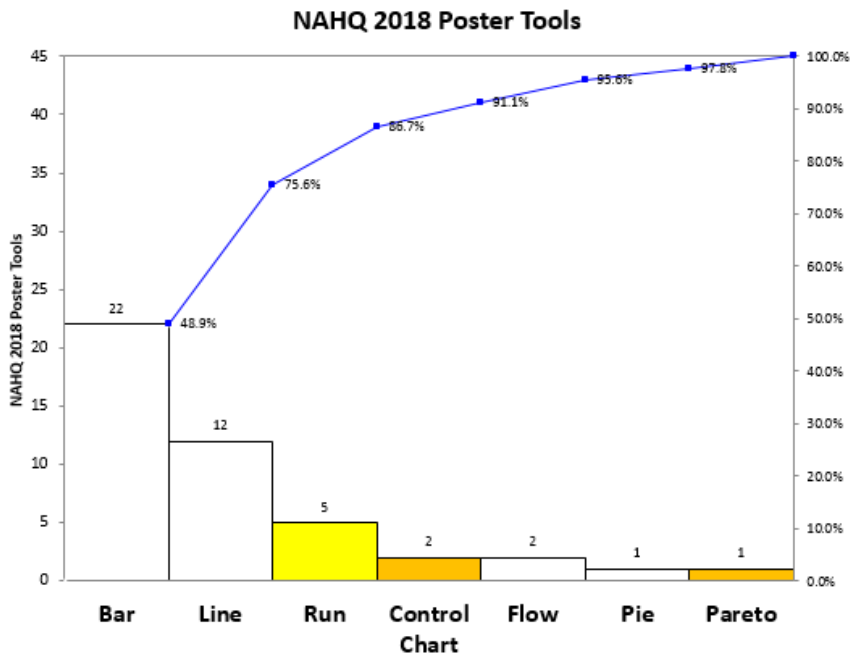
Hacking Six Sigma



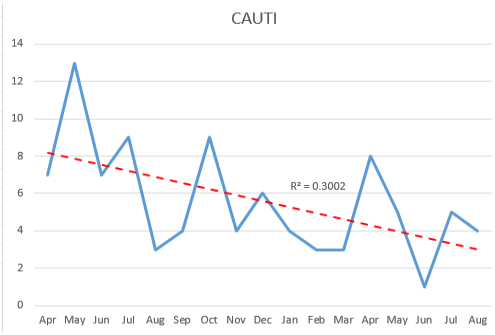
Make Sure Your Story Uses the Magnificent Seven Tools

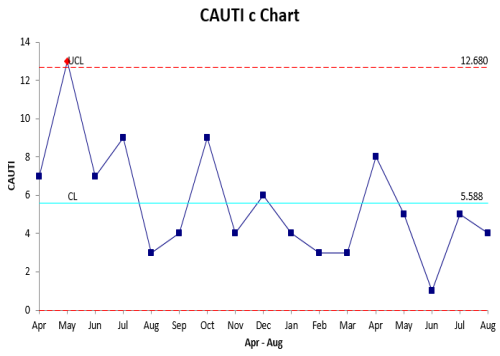

IHI 2017 Quality Tool Usage
IHI 2017 Quality Tool Usage
Tool Usage
Bar
Line
Control Chart
Fishbone
Flow Chart
Pie
Run Chart
Pareto Chart
Box Whisker
Other
101
75
41
31
28
27
21
14
3
6
29.1%
50.7%
62.5%
71.5%
79.5%
87.3%
93.4%
97.4%
98.3%
100.0%


Magnet 2018 Poster Tools
Magnet Poster Tools
Tools
Bar
line
pie
Pareto
Run Chart
Control Chart
Fishbone
Flow chart
Box Whisker
69
34
6
3
3
2
2
2
1
56.6%
84.4%
89.3%
91.8%
94.3%
95.9%
97.5%
99.2%
100.0%

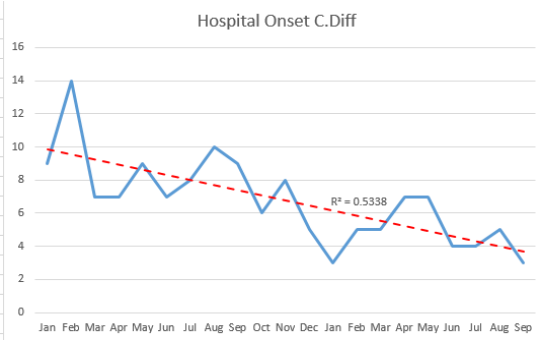


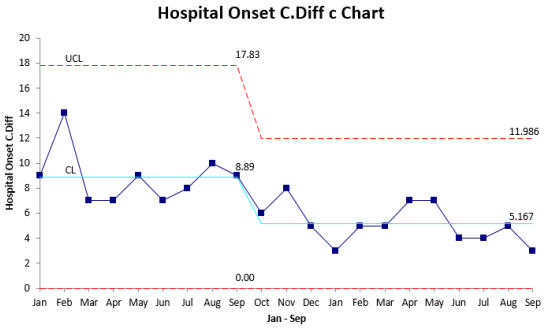
What's Wrong With Line Charts?

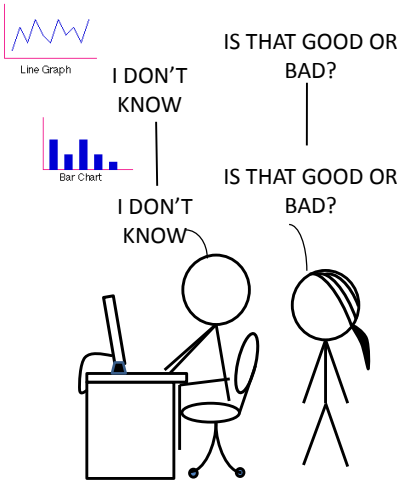


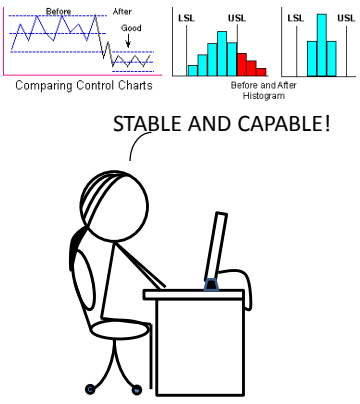


What's Wrong With Line Charts?











Why Raw Data?

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
1	Total	Deaths	CABG Surgeries	Percent		Hospital A	Deaths	CABG Surgeries	Percent		Hospital B	Deaths	CABG Surgeries	Percent		Hospital C	Deaths	CABG Surgeries	Percent		Hospital D	Deaths	CABG Surgeries	Percent
2	Q1	14	427	3.3%		Q1	4	199	2.0%		Q1	6	126	4.8%		Q1	4	88	4.5%		Q1	0	14	0.0%
3	Q2	12	430	2.8%		Q2	3	204	1.5%		Q2	2	121	1.7%		Q2	5	90	5.6%		Q2	2	15	13.3%
4	Q3	18	441	4.1%		Q3	5	194	2.6%		Q3	9	140	6.4%		Q3	3	89	3.4%		Q3	1	18	5.6%
5	Q4	12	478	2.5%		Q4	6	232	2.6%		Q4	3	123	2.4%		Q4	1	100	1.0%		Q4	2	23	8.7%
6	Q1	13	442	2.9%															1.0%		Q1	1	23	4.3%
7	Q2	7	402	1.7%															2.4%		Q2	0	11	0.0%
8	Q3	13	367	3.5%															3.9%		Q3	1	13	7.7%
9	Q4	11	414	2.7%															4.8%		Q4	2	20	10.0%
10	Q1	11	400	2.8%															4.3%		Q1	0	10	0.0%
11	Q2	13	439	3.0%		Q2	3	271	1.1%		Q2	7	89	7.9%		Q2	3	70	4.3%		Q2	0	9	0.0%
12	Q3	12	404	3.0%		Q3	7	250	2.8%		Q3	2	71	2.8%		Q3	2	73	2.7%		Q3	1	10	10.0%
13	Q4	5	414	1.2%		Q4	4	265	1.5%		Q4	0	69	0.0%		Q4	1	65	1.5%		Q4	0	15	0.0%
14	Q1	9	477	1.9%		Q1	3	298	1.0%		Q1	0	85	0.0%		Q1	3	74	4.1%		Q1	3	20	15.0%
15	Q2	10	440	2.3%		Q2	6	251	2.4%		Q2	2	100	2.0%		Q2	2	72	2.8%		Q2	0	17	0.0%
16	Q3	13	419	3.1%		Q3	6	258	2.3%		Q3	4	95	4.2%		Q3	2	52	3.8%		Q3	1	14	7.1%
17	Q4	17	431	3.9%		Q4	9	268	3.4%		Q4	5	68	7.4%		Q4	0	75	0.0%		Q4	3	20	15.0%

Because Summarized Data Has Little Nutritional Value



Raw Data Minimum Requirement

When – Where – What?

How Much? Who? Why?

If You Have Data Like This…

	A	B	C	D	E	F	G	H	I	J	K
1	Region	POST DATE	ENT	ADM DATE	DIS DATE	AS	COS	FC	IN1	PT	DENIED CHARGES
2	North	6/27/03	H								543.07
3	South	12/24/02	Hosp2	7/13/02	1/1/00	OL		X	BCP	E	215.4
4	South	2/25/03	Hosp2	12/6/02	1/1/00			Y	CGH	O	157.92
5	South	5/2							MAH	O	90.73
6	North	7/1							HEH	O	4103.78
7	North	11							PTB	E	3224.83
8	North	11/20/02	Hosp5	4/15/02	1/1/00	OL		F	PTB	O	3291.76
9	North	11/27/02	Hosp1	5/13/02	1/1/00	OL		F	PTB	O	13845.9
10	North	11/27/02	Hosp4	9/16/02	1/1/00			F	PTB	O	1151
11	North	12/11/02	Hosp6	9/24/02	1/1/00			F	PTB	O	797
12	North	6/18/03	Hosp7	4/15/03	1/1/00			C	OT1	O	215
13	North	6/25/03	Hosp5	7/26/01	1/1/00	OL		F	PTB	O	2124.86
14	North	8/29/03	Hosp4	6/2/03	1/1/00			F	PTB	O	12
15	North	9/5/03	Hosp6	8/8/03	1/1/00			F	PTB	O	304
16	North	9/5/03	Hosp6	8/8/03	1/1/00			F	PTB	O	89
17	North	9/5/03	Hosp6	8/8/03	1/1/00			F	PTB	O	89

Where When What How Much

$5,000,000/Year

If You Have Data Like This…

	A	B	C	D	E	F
1	PROC_DATE	COST_CTR	SCRAP_CODE	SCRAP_DESC	WEIGHT_IN	SCRAP_WGT
2	02/22/07	CC295	SB111	Tracking	434	122
3	02/21/07	CC290	WF4	STAINS	520	3,861
4	02/22/07	CC295	SB58	TEST	459	365
5	02/28/07	CC290	WF4	STAINS	2,780	2,780
6	02/28/07	CC290	T75	WORK RCVD DENTS	4,018	4,018
7	03/12/07	CC290	CF137	Stick	4,239	588
8	04/16/07	CC280	Q76	RESEARCH / DEVELOPMEN	6,526	6,526
9	04/06/07	CC290	SH93	CUSTOMER DOES NOT WAI	4,524	4,524
10	02/28/07	CC290	WF4	STAINS	4,375	925
11	01/23/07	CC280	PL132	Paint Line Set Up Samples	8,596	616
12	01/23/07	CC280	PL76	RESEARCH / DEVELOPMEN	8,596	2,800
13	02/12/07	CC295	SB68	FLOOR DAMAGE	5,180	123
14	03/12/07	CC290	CF137	Stick	2,924	272

Where When Where What

$1,000,000/Year

Data Mining Wizardry

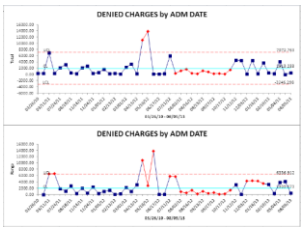
	A	B	C	D	E	F	G	H	I	J	K
1	Region	POST DATE	ENT	ADM DATE	DIS DATE	AS	COS	FC	IN1	PT	DENIED CHARGES
2	North	6/27/03	Hosp1	2/13/03	1/1/00	OL		X	AEH	O	543.07
3	South	12/24/02	Hosp2	7/13/02	1/1/00	OL		X	BCP	E	215.4
4	South	2/25/03	Hosp2	12/6/02	1/1/00			X	CGH	O	157.92
5	South	5/23/03	Hosp2	10/20/02	1/1/00	OL		X	MAH	O	90.73
6	North	7/15/0						X	HEH	O	4103.78
7	North	11/5/0						F			
8	North	11/20/0						F			
9	North	11/27/0						F			
10	North	11/27/0						F			
11	North	12/11/02	Hosp6	9/24/02	1/1/00			F			
12	North	6/18/03	Hosp7	4/15/03	1/1/00			C			

Raw Data

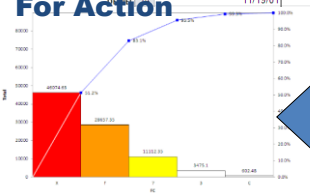
Pivot Tables

	A	B
1	Region	(All)
2		
3	Sum of DENIED CHARGES	ENT
4	ADM DATE	Hosp1 Hosp2 Hosp3 Hosp4 Hosp5 Hosp6 Hosp7 Grand Total
5		7.48
6		9.62
7		8.98
8		1.16
9		4.86
10		4.83
11		7.16
12		0.42
13		6.16
14	11/19/01	2627.84 2627.84

Counted & Categorized Data



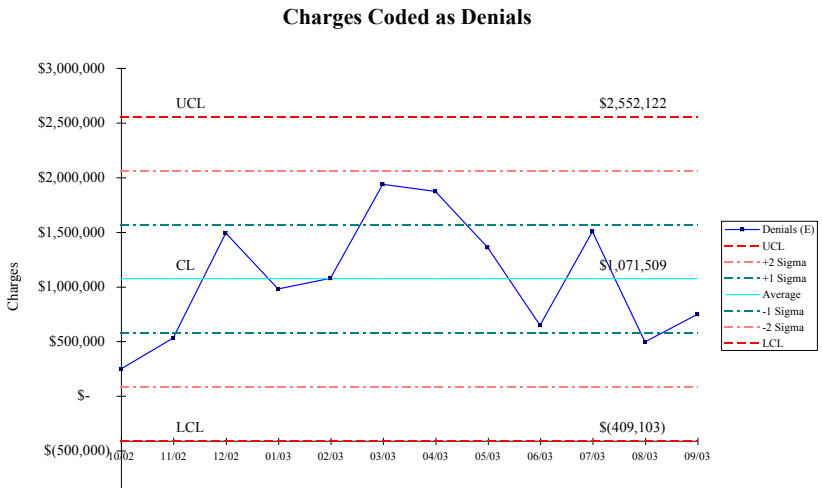
Insights For Action



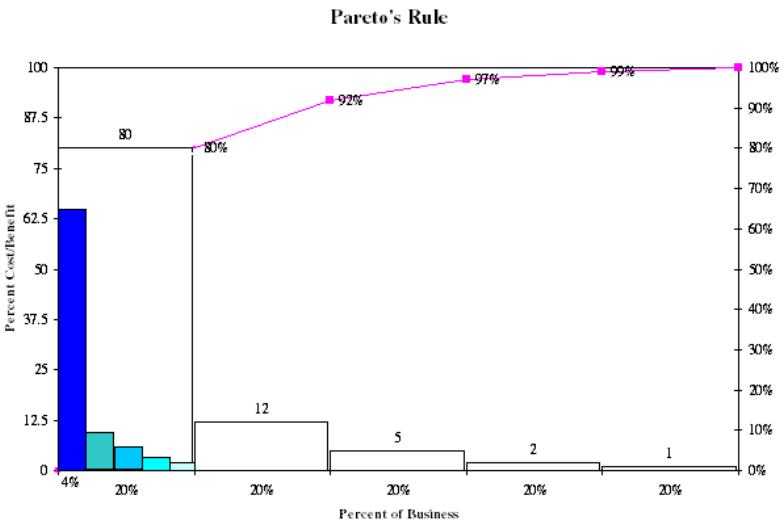
Select the Right Charts

Focus on Performance

Start with a control chart of defects, mistakes, errors or costs:

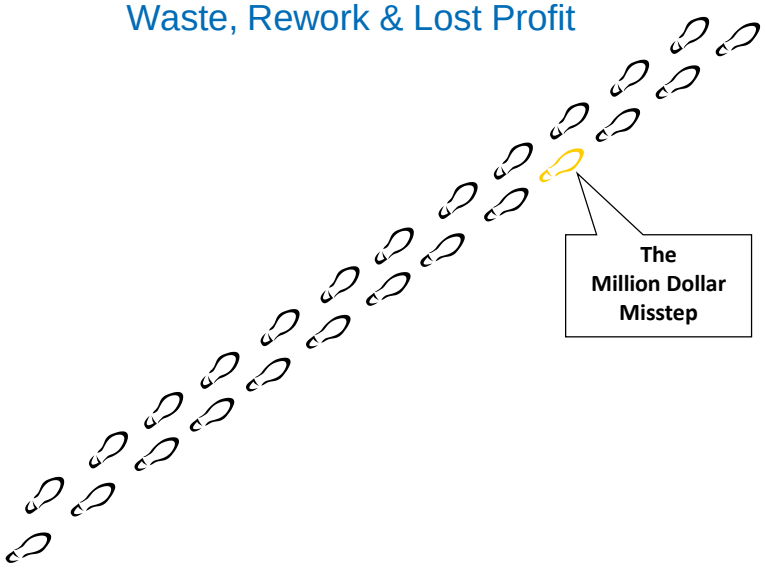


Go Deeper – Apply the 4-50 Rule

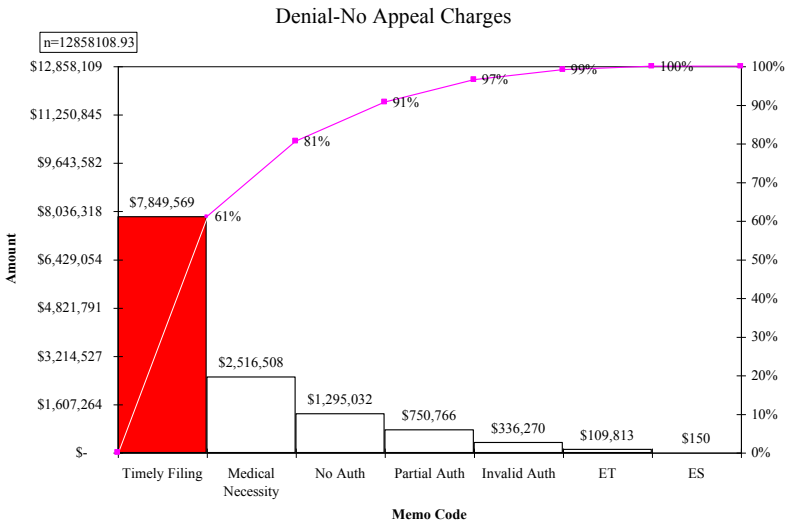


4% of the Process Results in over 50% of the Errors

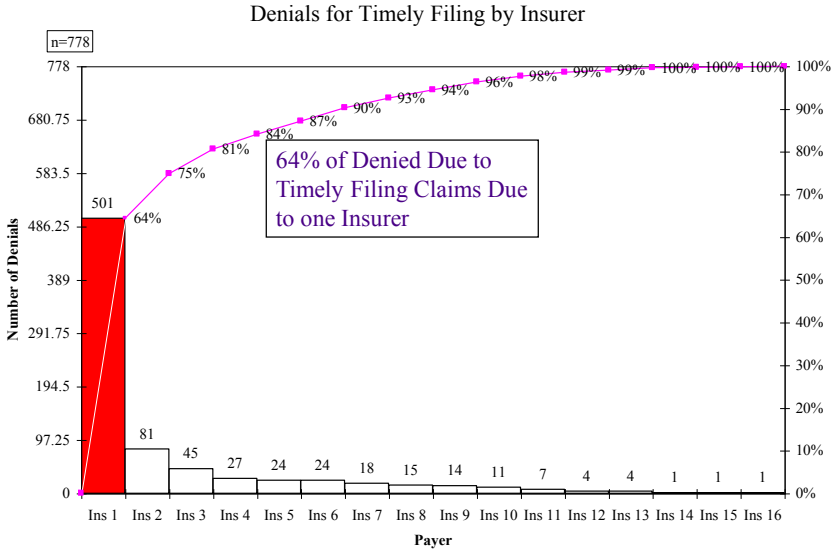
1 Step of 25 Creates 50% of the Waste, Rework & Lost Profit



Use Paretos to Find What to Fix

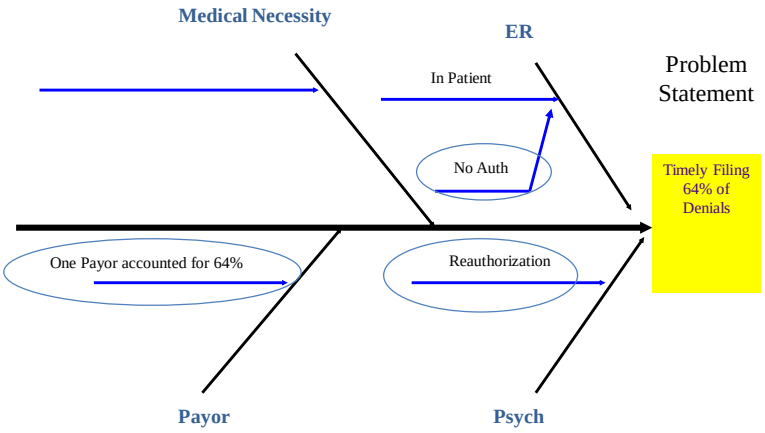


Use Paretos to Find What to Fix



Identify Root Causes

Use the fishbone diagram to perform root cause analysis:

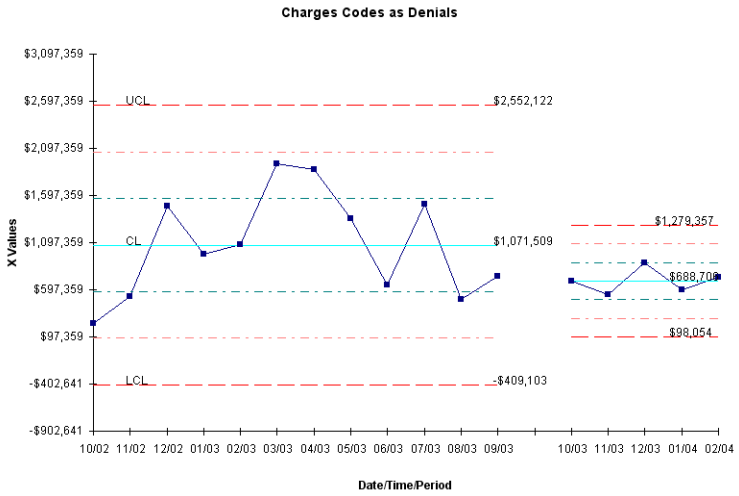


Countermeasures (Mistake Proofing)

Countermeasure

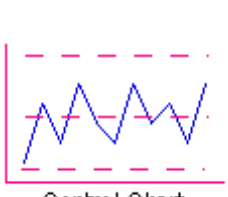
- Work with insurance company to resolve excessive denials
- Reduce delays that contribute to timely filling denials

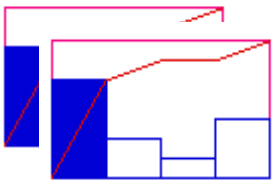
Verify Results

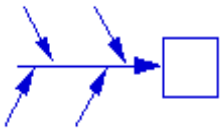


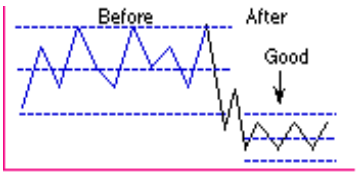
Reduced denials by 35% starting the following Monday ($380,000/month)

World Class Improvements

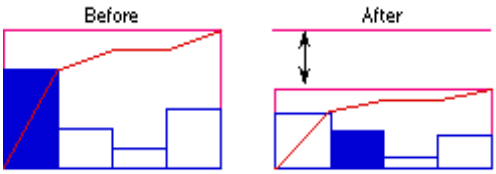

Control Chart


Pareto Chart


Cause-Effect Diagram

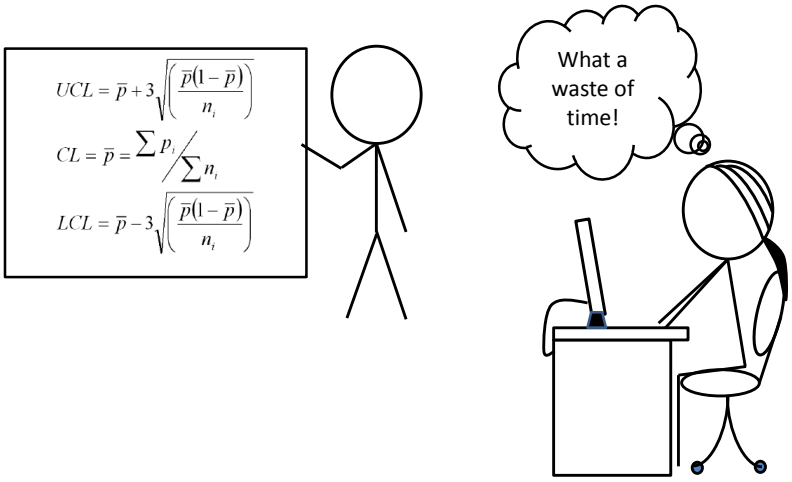


Comparing Control Charts

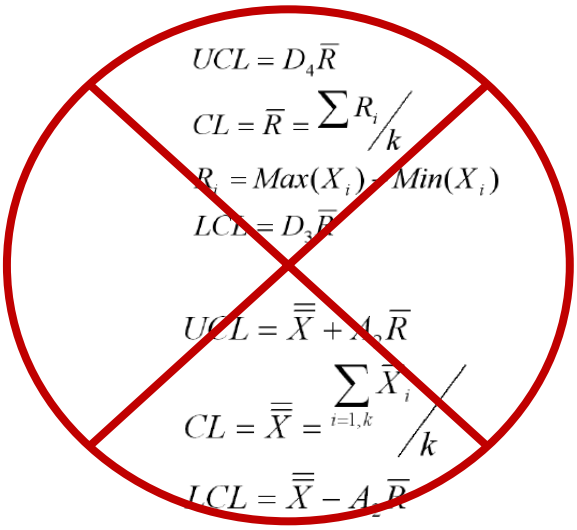


Comparing Pareto Charts

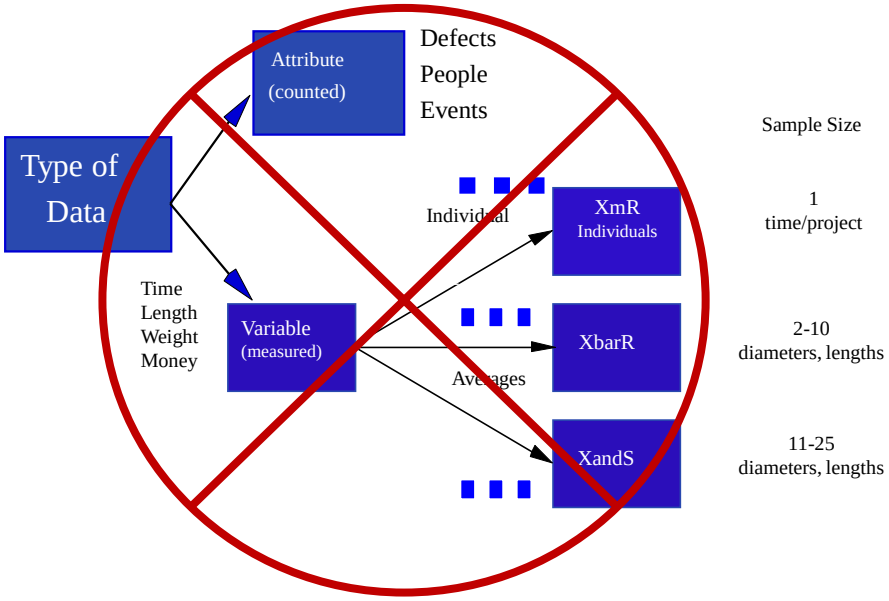
Hacking Six Sigma Training



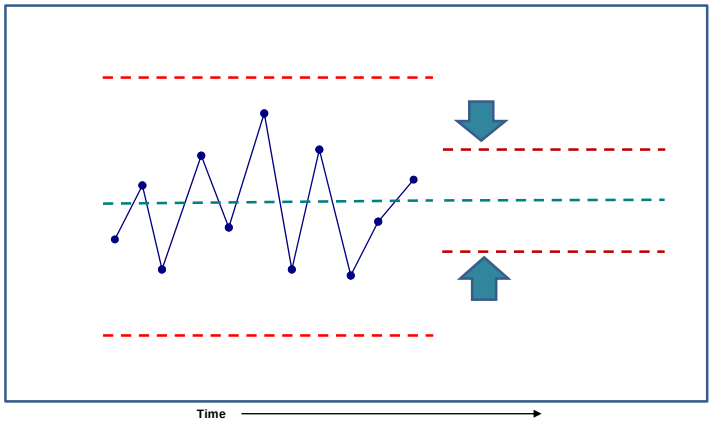
You Don't Need to Learn Formulas



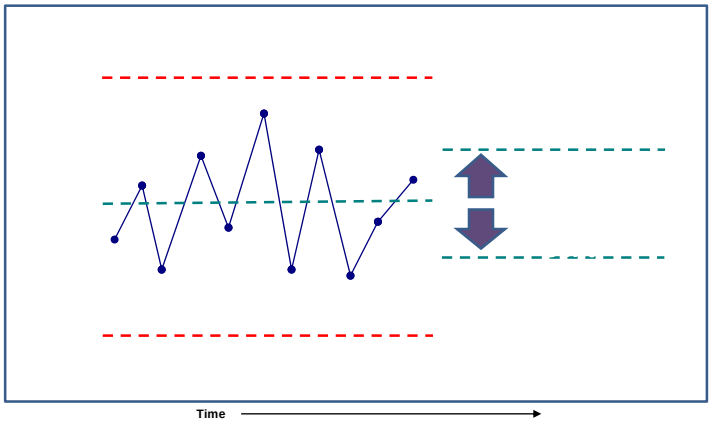
You Don't Need to Learn Decision Trees



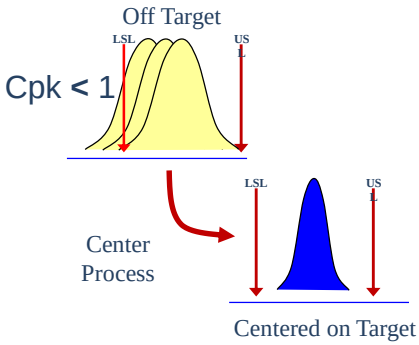
Learn the Patterns – Reduce Variation



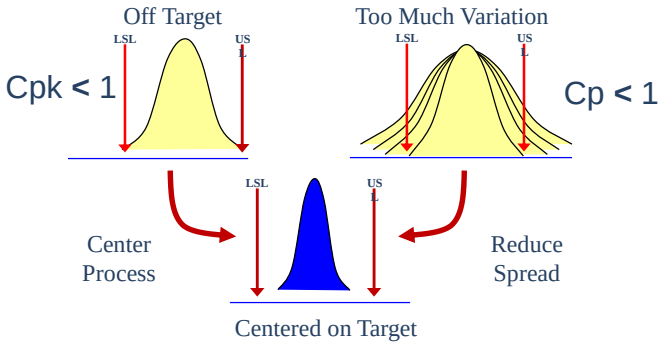
Move Center Line Down or Up



Reduce Deviation

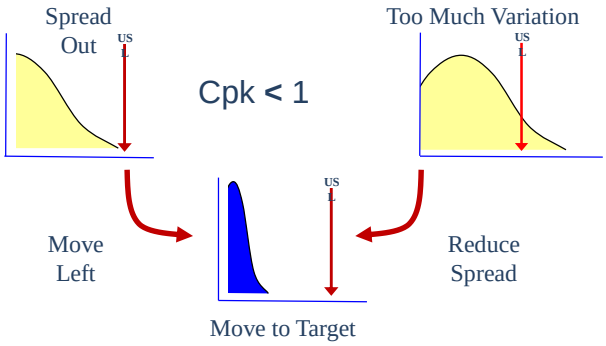


Reduce Deviation



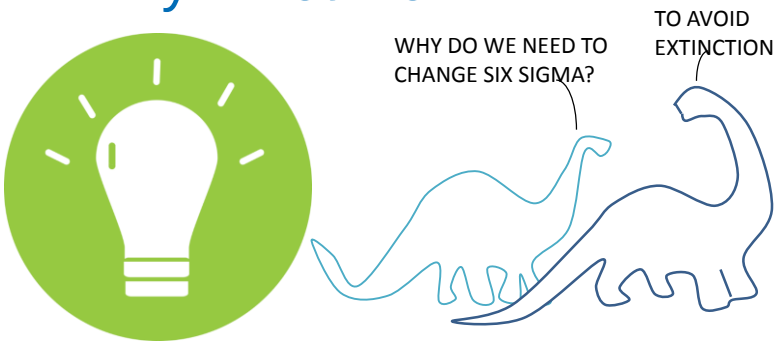
Cpk > 1

Cycle-Time Deviation



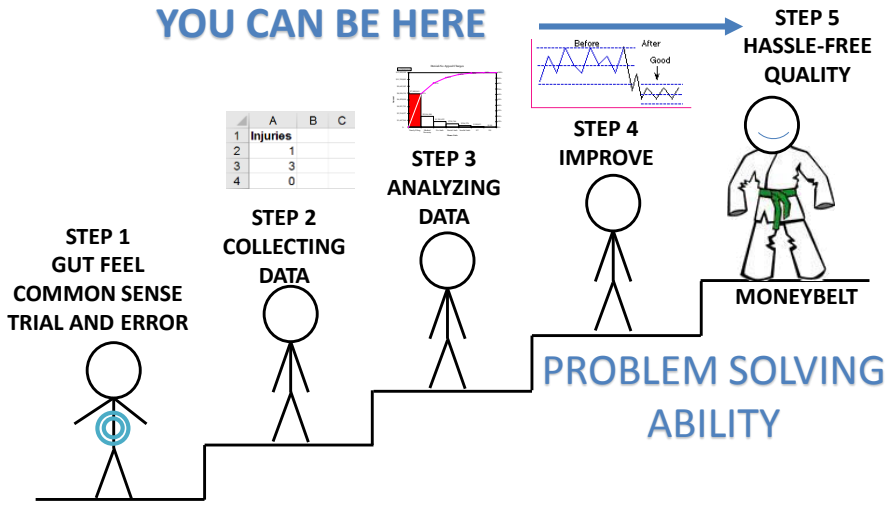
Cpk > 1

Why Pivot Now?



Agile Lean Six Sigma

Traditional	Agile
• Big Implementations • Rigid processes • Maintaining the status quo • Big projects • Silos • Complexity	• Small Agile Teams • Agile • Accelerating speed and adaptability • Laser-focused, data-driven projects • Engagement • Simplicity



Take-aways

- Discovered how one-day Lean Six Sigma Yellow Belt training can accelerate the implementation of Lean Six Sigma and deliver results.
- Learned how to identify the best candidates for Green or Black Belt training by using an Agile approach.
- Discovered how companies are using Agile to succeed where tradition implementations have failed.
- Identified the seven key tools for Lean Six Sigma success.
- Learned how to teach Lean Six Sigma in one day and generate results.

Questions?

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QI Macros – Affordable Tools for the Statistically Challenged



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